

Chapter 3.2 Control Charts

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Plan

- 1 Univariate Control Chart
 - Univariate Control Chart for a Mean
 - Univariate Control Chart for a Standard Deviation
- 2 Multivariate Control Charts

Univariate Control Chart for a Mean

- 1 Observe m values of x at each of n Time Points: x_{ik} ,
 $i = 1, \dots, n$; $k = 1, \dots, m$
- 2 Obtain means and standard deviations : $\bar{x}_1, \dots, \bar{x}_n$; $s_1, \dots, s_n$:

$$\bar{x}_i = \frac{1}{m} \sum_{k=1}^m x_{ik} \quad s_i = \sqrt{\frac{1}{m-1} \sum_{k=1}^m (x_{ik} - \bar{x}_i)^2} \quad i = 1, \dots, n$$

- 3 Obtain the overall mean and standard deviation:

$$\bar{\bar{x}} = \frac{1}{n} \sum_{i=1}^n \bar{x}_i \quad s = \sqrt{\frac{1}{n(m-1)} \sum_{i=1}^n (m-1) s_i^2}$$

- 4 Obtain Upper and Lower Control Limits (Overall Mean + / - 3SE):
 $UCL = \bar{\bar{x}} + 3 \frac{s}{\sqrt{m}} \quad LCL = \bar{\bar{x}} - 3 \frac{s}{\sqrt{m}}$

- 5 Plot the Observed
 $\bar{x}_i$ versus i with points joined by lines and Add Lines for LCL , $\bar{\bar{x}}$, UCL
- 6 Points outside of (LCL, UCL) are considered out-of-control

Univariate Control Chart for a Mean II

- 7. If Unequal Sample Sizes:

$$\begin{aligned}\bar{x}_i &= \frac{1}{m_i} \sum_{k=1}^{m_i} x_{ik} & s_i &= \sqrt{\frac{1}{m_i-1} \sum_{k=1}^{m_i} (x_{ik} - \bar{x}_i)^2} \\ \bar{\bar{x}} &= \frac{\sum_{i=1}^n m_i \bar{x}_i}{\sum_{i=1}^n m_i} & s &= \sqrt{\frac{\sum_{i=1}^n (m_i-1) s_i^2}{\sum_{i=1}^n (m_i-1)}} \\ UCL_i &= \bar{\bar{x}} + 3 \frac{s}{\sqrt{m_i}} & LCL_i &= \bar{\bar{x}} - 3 \frac{s}{\sqrt{m_i}}\end{aligned}$$

Example – NFL Home Team Point Spread Deviation by Week – 2018

Data:

- Point Spread for Home Team (Negative means Home Team Favored) $\equiv$ PS
- Point Differential for Home Team (Home Points - Visitor Points) $\equiv$ HD
- Spread Differential (HD + PS, Positive means Home Team Covered the Spread)

Week (i)	m_i	xbar_i	s_i
1	16	-3.719	15.695
2	16	0.781	9.822
3	16	0.500	13.858
4	15	3.233	12.865
5	15	5.900	7.124
6	15	-2.500	15.183
7	14	-4.607	15.754
8	14	-0.714	9.147
9	13	-1.462	16.056
10	14	-2.286	19.463
11	13	3.423	12.185
12	15	3.033	8.776
13	16	2.250	9.798
14	16	1.469	11.051
15	16	1.875	11.100
16	16	-3.594	7.035
17	16	-5.188	15.937

$$\bar{\bar{x}} = \frac{16(-3.719) + \cdots + 16(-5.188)}{16 + \cdots + 16} = \frac{-26.00}{256} = -0.102$$

$$s = \sqrt{\frac{(16-1)15.695^2 + \cdots + (16-1)15.937^2}{(16-1) + \cdots + (16-1)}} = \sqrt{\frac{39141.95}{239}} = 12.797$$

Typically, the Center Line is set at $\bar{\bar{x}}$ and

$$LCL_i = \bar{\bar{x}} - 3\frac{s}{\sqrt{m_i}} \quad \text{and} \quad UCL_i = \bar{\bar{x}} + 3\frac{s}{\sqrt{m_i}}$$

Here target Mean is 0, so that is center line and

$$LCL_i = -3\frac{s}{\sqrt{m_i}} \quad UCL_i = +3\frac{s}{\sqrt{m_i}}$$



The “Spread Differential Process” varies, but never becomes “out of control”

Univariate Control Chart for a Standard Deviation

$$X_1, \dots, X_n \overset{i.i.d}{\sim} \mathcal{N}(\mu, \sigma^2)$$

$$\Rightarrow E\{S^2\} = \sigma^2 \text{ but } E\{S\} = k\sigma \quad V\{S\} = \sigma^2 - (k\sigma)^2 = \sigma^2(1 - k^2)$$

$$\Rightarrow \hat{SE}\{S\} = \frac{S}{k} \sqrt{1 - k^2}.$$

$$\text{Average Standard Deviation: } \bar{S} = \sqrt{\frac{\sum_{i=1}^n (m_i - 1) S_i^2}{\sum_{i=1}^n (m_i - 1)}}.$$

$$\text{Center - Line} = \bar{S} \quad LCL_i = \bar{S} - 3 \frac{\bar{S}}{k_i} \sqrt{1 - k_i^2} \quad UCL_i = \bar{S} + 3 \frac{\bar{S}}{k_i} \sqrt{1 - k_i^2}$$

$$\text{with } k_i = \sqrt{\frac{2}{m_i - 1}} \left(\frac{\Gamma(\frac{m_i}{2})}{\Gamma(\frac{m_i - 1}{2})} \right).$$

Example – NFL Home Team Point Spread Deviation by Week – 2018

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17	16	-5.188	15.937

Example – NFL Home Team Point Spread Deviation by Week – 2018

$$\bar{S} = \sqrt{\frac{(16-1)15.695^2 + \dots + (16-1)15.937^2}{(16-1) + \dots + (16-1)}} = \sqrt{\frac{39141.95}{239}} = 12.797$$

$$k_1 = \sqrt{\frac{2}{16-1}} \left(\frac{\Gamma\left(\frac{16}{2}\right)}{\Gamma\left(\frac{16-1}{2}\right)} \right) = \sqrt{\frac{2}{15}} \left(\frac{5040}{1871.254} \right) = 0.9835$$

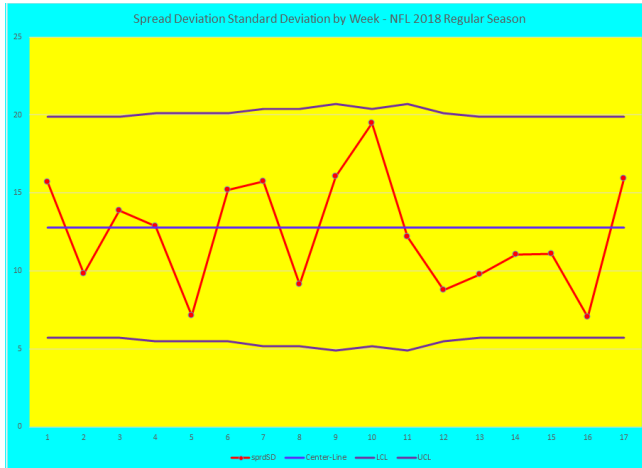
$$LCL_i = \bar{S} - 3 \frac{\bar{S}}{k_i} \sqrt{1 - k_i^2}$$

$$\Rightarrow LCL_1 = 12.797 - 3 \left(\frac{12.797}{0.9835} \right) \sqrt{1 - 0.9835^2} = 12.797 - 7.062 = 5.735$$

$$UCL_i = \bar{S} + 3 \frac{\bar{S}}{k_i} \sqrt{1 - k_i^2}$$

$$\Rightarrow UCL_1 = 12.797 + 3 \left(\frac{12.797}{0.9836} \right) \sqrt{1 - 0.9836^2} = 12.797 + 7.062 = 19.859$$

Example – NFL Home Team Point Spread Deviation by Week – 2018



The standard deviation varies, but never gets “out of control”

Plan

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Multivariate Control Charts with 1 Observation per Time Point

$$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ with } \mathbf{X}_j = \begin{bmatrix} X_{j1} \\ \vdots \\ X_{jp} \end{bmatrix} \quad \bar{\mathbf{X}} = \frac{1}{n} \sum_{j=1}^n \mathbf{X}_j$$

$$E\{\mathbf{X}_j\} = \boldsymbol{\mu} \quad V\{\mathbf{X}_j\} = \boldsymbol{\Sigma} \quad E\{\bar{\mathbf{X}}\} = \boldsymbol{\mu} \quad V\{\bar{\mathbf{X}}\} = \frac{1}{n}\boldsymbol{\Sigma} \quad \text{COV}\{\mathbf{X}_j, \bar{\mathbf{X}}\} = \frac{1}{n}\boldsymbol{\Sigma}$$

$$\Rightarrow E\{\mathbf{X}_j - \bar{\mathbf{X}}\} = \boldsymbol{\mu} - \boldsymbol{\mu} = \mathbf{0} \quad V\{\mathbf{X}_j - \bar{\mathbf{X}}\} = \boldsymbol{\Sigma} + \frac{1}{n}\boldsymbol{\Sigma} - 2\frac{1}{n}\boldsymbol{\Sigma} = \frac{n-1}{n}\boldsymbol{\Sigma}$$

$$\mathbf{X}_j - \bar{\mathbf{X}} \sim \mathcal{N}\left(\mathbf{0}, \frac{n-1}{n}\boldsymbol{\Sigma}\right) \text{ but } \mathbf{X}_j - \bar{\mathbf{X}} \text{ and } \mathbf{S} \text{ are not independent}$$

Assume an approximate chi-square distribution for $(\mathbf{X}_j - \bar{\mathbf{X}})' \mathbf{S}^{-1} (\mathbf{X}_j - \bar{\mathbf{X}})$

After observing a sample $\mathbf{x}_1, \dots, \mathbf{x}_n$ construct approximate $(1 - \alpha)$ quality ellipse:

$\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) \leq \chi_p^2(\alpha)$ Observations $\mathbf{x}_j$ outside of ellipse are "out-of-control"

T^2 -Chart : Plot $T_j^2 = (\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}})$ versus j with $UCL = \chi_p^2(\alpha)$

Example – Wilt Chamberlain's 1961-2 Regular Season

$x_{j1} \equiv$ Rebounds in game j $x_{j2} \equiv$ Points in game j $n = 80$ games

$$\bar{\mathbf{x}} = \begin{bmatrix} 25.65 \\ 50.36 \end{bmatrix} \quad \mathbf{S} = \begin{bmatrix} 34.21 & 13.90 \\ 13.90 & 143.88 \end{bmatrix} \quad \mathbf{\Lambda} = \begin{bmatrix} 146.6145 & 0 \\ 0 & 32.4705 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 0.123814 & -0.992306 \\ 0.992306 & 0.123814 \end{bmatrix}$$

Approximate $(1 - \alpha)$ 100% Quality Ellipse:

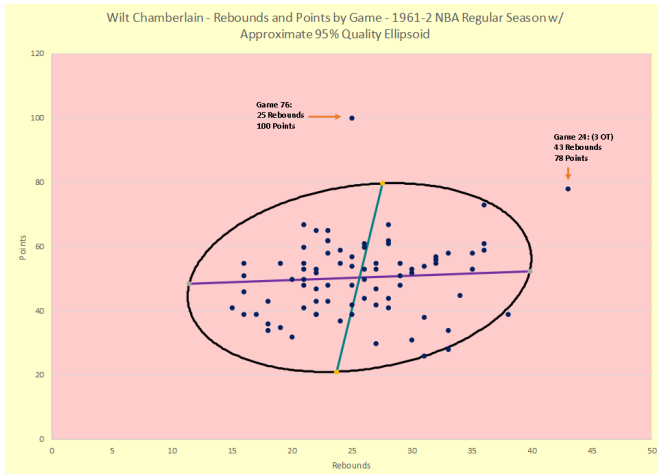
$$\mathbf{x} : (\mathbf{x} - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) \leq \chi_p^2(\alpha)$$

$$\Rightarrow \mathbf{x} : \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 25.65 \\ 50.36 \end{bmatrix} \right)' \begin{bmatrix} 34.21 & 13.90 \\ 13.90 & 143.88 \end{bmatrix}^{-1} \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 25.65 \\ 50.36 \end{bmatrix} \right) \leq 5.991$$

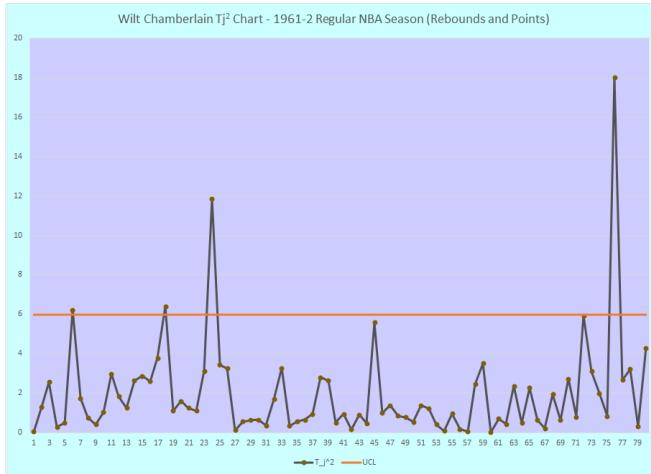
$$(p = 2, \alpha = 0.05)$$

$$T_j^2 = (\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{S}^{-1} (\mathbf{x}_j - \bar{\mathbf{x}}) = \left(\begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix} - \begin{bmatrix} 25.65 \\ 50.36 \end{bmatrix} \right)' \begin{bmatrix} 34.21 & 13.90 \\ 13.90 & 143.88 \end{bmatrix}^{-1} \left(\begin{bmatrix} x_{j1} \\ x_{j2} \end{bmatrix} - \begin{bmatrix} 25.65 \\ 50.36 \end{bmatrix} \right)$$

$$UCL = 5.991 \quad (p = 2, \alpha = 0.05)$$



Note that 4/80 games fall outside the 95% Confidence ellipsoid, with 2 very extreme games (one with respect to points and not rebounds, the other with respect to both)



Games 24 (43 Rebounds, 78 Points, 3 OTs) and 76 (25 Rebounds, 100 Points) are “out of control” – especially for the opposing team

Multivariate Control Chart for Process Variation

- 1 Procedure based on the Generalized Inverse of the Variance - Covariance Matrix: $|\mathbf{S}|$
- 2 Obtain m_i measurements for p variables in period i , $i = 1, \dots, n$
- 3 Compute the Covariance matrix for period i : $\mathbf{S}_i$ $i = 1, \dots, n$

- 4 Obtain the Average of the Covariance matrices: $\mathbf{S} = \frac{\sum_{i=1}^n (m_i - 1) \mathbf{S}_i}{\sum_{i=1}^n (m_i - 1)}$

$$E\{|\mathbf{S}_i|\} = \left[\frac{1}{(m_i - 1)^p} \prod_{k=1}^p (m_i - k) \right] |\mathbf{\Sigma}| = k_{1i} |\mathbf{\Sigma}|$$

$$V\{|\mathbf{S}_i|\} = \left[\frac{1}{(m_i - 1)^{2p}} \prod_{k=1}^p (m_i - k) \left[\prod_{l=1}^p (m_i - l + 2) - \prod_{l=1}^p (m_i - l) \right] \right] |\mathbf{\Sigma}|^2 = k_{2i} |\mathbf{\Sigma}|^2$$

These lead to Choices of Center - Line and Upper and Lower Control Limits ($\pm 3SD$) :

$$CL_i = |\mathbf{S}| \quad UCL_i = \frac{|\mathbf{S}|}{k_{1i}} \left(k_{1i} + 3\sqrt{k_{2i}} \right) \quad LCL_i = \max \left(\frac{|\mathbf{S}|}{k_{1i}} \left(k_{1i} - 3\sqrt{k_{2i}} \right), 0 \right)$$

Example – NFL Point Spread and O/U Differentials by Week - 2018

$$X_1 \equiv \text{Home Spread Differential} = \text{Home Score} - \text{Opp Score} + \text{Spread}$$

$$X_2 \equiv \text{Over/Under Differential} = \text{Home Score} + \text{Opp Score} - \text{O/U}$$

Week (i)	m_i	s_11i	s_22i	s_12i	S_i
1	16	246.35	205.15	-73.39	45151.58
2	16	96.48	144.69	15.99	13704.37
3	16	192.03	146.33	4.07	28084.34
4	15	165.51	201.87	-21.36	32954.63
5	15	50.75	150.73	-9.99	7549.28
6	15	230.52	196.47	90.20	37155.05
7	14	248.19	88.46	-31.16	20982.89
8	14	83.67	124.96	36.93	9091.39
9	13	257.79	168.95	-43.37	41672.47
10	14	378.82	230.80	8.52	87358.75
11	13	148.47	183.74	-18.68	26930.70
12	15	77.02	100.76	-10.22	7655.42
13	16	96.00	219.96	-18.09	20789.29
14	16	122.11	166.01	-35.93	18982.12
15	16	123.20	136.09	18.08	16439.79
16	16	49.50	186.81	36.80	7892.46
17	16	253.99	282.98	49.61	69413.69

Example – NFL Point Spread and O/U Differentials by Week - 2018

$$|\mathbf{S}| = \frac{(16-1)(45151.58) + \dots + (16-1)(69413.69)}{(16-1) + \dots + (16-1)} = \frac{6851133.33}{239} = 28655.83$$

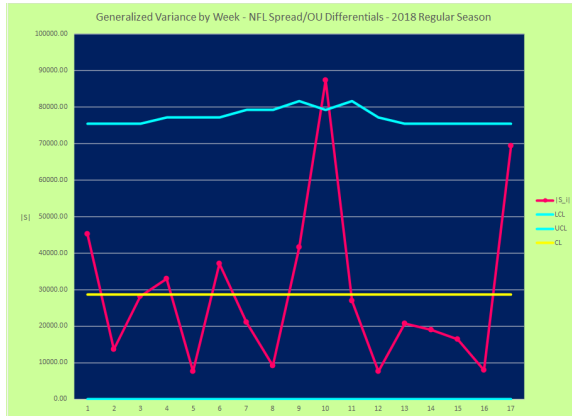
$$k_{11} = \frac{(16-1)(16-2)}{(16-1)^2} = 0.9333$$

$$k_{21} = \frac{(16-1)(16-2)}{(16-1)^2} [(16+1)(16) - (16-1)(16-2)] = 0.2572$$

$$LCL_1 = \max \left(0, 28655.83 \left[\frac{0.9333 - 3\sqrt{0.2572}}{0.9333} \right] \right) = \max(0, -18064.62) = 0$$

$$UCL_1 = 28655.83 \left[\frac{0.9333 + 3\sqrt{0.2572}}{0.9333} \right] = 75393.27$$

Example – NFL Point Spread and O/U Differentials by Week - 2018



Week 10 had an exceptionally large Generalized Variance (and Week 17 was large as well)

Example – NFL Point Spread and O/U Differentials by Week - 2018

S	
163.77	0.33
0.33	173.33

S_10	
378.82	8.52
8.52	230.80

S_17	
253.99	49.61
49.61	282.98